

Evaluation of Web-Based Social Network Extraction

The document explores the complex relationship between real-world social networks and their online reflections, particularly focusing on data extracted from the Web.

1. Background and Contrasting Views The Internet and computer communication networks have been studied as indicators for social networks since at least 1996. Some studies suggest that internet interaction replaces traditional social networks [1, OR02, WSD+96].

- **Physics/Computer Science Perspective:** Researchers from these backgrounds often view online networks as direct equivalents of real-world networks.
- **Social Science Perspective:** Network analysts with a social science background tend to be more cautious, often treating electronic communication networks and online communities as a separate field requiring different theories and methods.

2. Core Research Question Despite extensive study, "surprisingly little is known about the exact relationship between real world networks and their online reflections". The central question is: To what extent does electronic data from the Web reveal social structures similar to those identified through traditional network questionnaires?

3. Focus of the Study This study specifically limits its attention to data generated by **social network mining from the Web based on co-occurrence**. This method is considered interesting because it can potentially reveal ties of different strengths based on the variety of underlying data. The authors suggest that electronic data like email traffic, which represents direct message exchanges, might have a higher correspondence with social network data compared to others like web logs, where a more distant perspective may be warranted.

4. Choice of Research Subject The study chose members of a **research organization** as its subjects. Scientific communities are particularly well-suited for online study due to the abundance and variety of available online information, such as personal homepages, research collaborations (projects, publications), and event descriptions (conferences, workshops).

5. Comparison to Survey Methods (The "Golden Standard") The study evaluates electronic data extraction against **survey methods**, which are the dominant approach for data collection in network analysis.

- **Advantages of Surveys:** Standard questionnaires are preferred in theory testing because their fixed structure ensures comparability of results across subjects and studies. They allow direct extraction of relational data and attributes for network analysis tools and statistical packages. Surveys are also minimally invasive and can be easily administered via mail, email, or online.
- **Limitations of Treating Surveys as "Golden Standard":** While dominant, network questionnaires are "hardly infallible". Any discrepancy found between online and offline analysis could be attributed to limitations of either method. Treating survey results as the "golden standard" means the study might inadvertently try to match their imperfections.

6. Contributions of the Study This study makes several unique contributions:

- It describes how an **evaluation of network extraction methods using real-world data** can be carried out, addressing a gap in existing knowledge.
- It uses this evaluation to **compare different network mining methods**, fine-tune their parameters, and measure the effect of disambiguation techniques.
- It provides an assessment of the quality of extraction methods by comparing **network measures** computed on web-based networks with those calculated from survey graphs.

7. Relevance and Advantages of Web-Based Extraction Web-based extraction is relevant for anyone considering using Web data as a proxy for real-world data because it requires significantly less effort than surveys or observation.

- **Scalability:** Results can be scaled to any number of participants.
- **Repeatability:** Data collection can be repeated multiple times, even retrospectively, if the community had a significant web presence.
- **Future Trend:** As more communities move online, web mining is expected to become a preferred data source for network experiments.

8. Data Availability Limitations The study's dataset cannot be made public due to privacy restrictions. This is a common limitation in real-world network analysis experiments, highlighting a general lack of publicly available benchmark datasets for evaluating network analysis methods transparently.

7.1 Differences between Survey Methods and Electronic Data Extraction

The document details potential sources of disagreement between results obtained from typical network surveys and those from Web mining.

1. Differences in What is Measured

- **Scope Limitation and Bias:** "What is not on the Web cannot be extracted". Web data can be biased if parts of a community are better represented online than others, similar to sampling problems in surveys.
- **Multiplex Networks:** Web extraction often finds evidence for different kinds of relationships, resulting in a **multiplex network** (a network where multiple types of relationships exist between the same set of actors). Disentangling these relationships is challenging, though machine learning can help by identifying features characterizing specific relationships (e.g., distinguishing co-authors from colleagues) [11, MHT+06]. The study addresses this by measuring multiple relationships in the survey to understand the composition of Web-found relationships.
- **Survey Question Formulation:** The equivalent challenge in surveys is precisely formulating questions to capture the desired relationship. The example of the General Social Survey (GSS) shows how a minor change in phrasing (from "personal matters" to "important matters") can significantly complicate comparisons due to varied respondent interpretation [12, 13, MSLB06, Rua98].

2. Errors Introduced by the Extraction Method These errors affect the accuracy of data extracted from the Web.

- **Homonymy:** Occurs with common names (e.g., J. Smith) where multiple individuals share the same name. This can be reduced by adding disambiguation terms to queries.
- **Synonymy:** Occurs when a person uses different variations of their name (e.g., James Hendler vs. Jim Hendler, different spellings, accents, alphabets). The study plans to experiment with measures to predict problematic names based on their extractability.
- **Systemic Errors/Noise:** Affect all names with equal probability, such as accidental co-occurrence of names on a webpage or duplicate co-occurrences. These errors are less threatening as they affect all cases uniformly and are likely reduced by large numbers of data points.

3. Errors Introduced by Survey Data Collection

- **Incomplete Coverage (Response Rates):** Surveys rarely cover a network completely. A response rate below 100% necessitates proof that non-respondents aren't significantly different or that results are robust enough to withstand non-response bias.

- **Respondent Cooperativeness and Fatigue:** Respondents vary in their willingness to cooperate and may experience fatigue, affecting data quality. These factors can be measured to assess their impact [16, MSLB06]. In small studies, non-respondents and respondents might cluster, influencing each other.
- **Observation Bias:** The act of observing itself can introduce bias, ideally affecting all subjects equally.
- **Question Formulation and Recall:** The type of relationship considered and the recall of contacts are influenced by how questions are formulated and their sequence in the survey [17, MSLB06].

7.2 Context of the Empirical Study

The study was conducted in September 2006, focusing on the social networks of **123 researchers** at the Department of Computer Science of the Vrije Universiteit, Amsterdam.

1. Department Structure The department was organized into six sections of varying sizes: Computer Systems (38), Artificial Intelligence (33), Information Management and Software Engineering (22), Business Informatics (17), Theoretical Computer Science (9), and Bioinformatics (4). These sections are further divided into groups, each led by a professor. The study included PhD students, postdocs, associate, and full professors, but excluded master students, support staff, and administrative support.

2. Nature of the Research Organization Unlike a broad research community (like the Semantic Web community), this organization is primarily bound by shared affiliation with an institution. However, this affiliation is rather loose, with the department level often seen as purely administrative. The overall structure resembles a research community, where relationships are largely driven by specific research interests or affiliation with smaller groups rather than the overall organization. The network of this organization was found to be even less centralized than the Semantic Web research community.

3. Reasons for Choosing this Community The author was a member of the Business Informatics group within the department, which allowed for direct interaction with participants and likely resulted in a higher response rate.

- **Response Rate:** Of 114 eligible researchers (after 9 opted out), 79 responded, yielding a **64% response rate**. The BI group (88%) and the AI group (79%) had above-average response rates.
- **Downside:** Some participants were uncomfortable providing personal information to a colleague, even with assurances of data anonymity and non-management use.

7.3 Data Collection

The study used a **custom-built online survey system** for data collection.

1. Advantages of Online Surveys

- **Accessibility:** Participants could log on anytime and did not need to be physically present.
- **Flexibility in Design:** Allowed adaptive questions based on previous answers, reducing completion time and non-response.
- **Easier Processing:** Electronic data was recorded directly in **RDF** (Resource Description Framework) using FOAF-based semantic representations. This meant no post-processing was needed for SNA (Social Network Analysis) packages, and the system performed real-time error checking.
- **Contrast with Off-the-Shelf Tools:** Commercial survey tools offer little customization for error checking or survey logic and typically produce difficult-to-process data formats like Excel sheets.

2. Survey Structure and Questions

The survey was divided into several pages:

- **Page 1:** Basic personal information (full/part-time status, age, years at organization, supervisor, research interests).
- **Pages 2-3:** Standard questions for self-monitoring and organizational identification (control variables, not used in this evaluation).
- **Page 4:** Participants selected people they knew from a complete list of Department members to pre-select potential contacts.
- **Page 5:** Participants specified the **nature of the relationship** with selected individuals.

3. Types of Relationships Surveyed

Six types of relationships were surveyed:

- **Instrumental Ties:**
 - **Advice Seeking:** "who you regularly approach if you have a work-related problem or when you want advice on a decision you have to make" [25, Kra90, IA93].
 - **Advice Giving:** "who regularly approaches you [for work-related problems or advice]".
 - **Future Cooperation:** "who would you like to work with in the future".
- **Affective Ties and General Perceptions:**
 - **Similarity Perceptions:** "who you consider especially similar to yourself" [26, MKB98].

- **Friendship:** "whom you consider a personal friend, e.g., a person you like to spend breaks with or engage in social activities" [26, MKB01, vdB99].
- **Adversarial Ties (Troubled Relation):** "with whom you have a difficult relation, e.g., you cannot get along with this person" [26, BBJ97, vdB99].

4. Data Collection Method: Roster Method The study used the **roster method** (providing a list of all potential contacts) instead of free recall (where participants name contacts themselves) [27, WFIG94].

- **Advantage of Roster Method:** Reduces the risk of participants forgetting to mention contacts (a problem with free recall where people name only those who first come to mind).
- **Measurement Error Mitigation:** Respondents were not restricted to a fixed number of nominations [27, HL73].
- **Two-Step Pruning Process:** The process of first pruning the list of participants before presenting relationship choices reduced strain on participants. This avoided large, sparse matrices common in traditional questionnaires, which increase errors, time, and abandonment rates. Participants found the survey "very smooth and easy to click through," taking about 15 minutes.

5. Data Storage Upon completion, the survey software stored results in a **Sesame RDF store**, saved backups, and notified the administrator via email [28, BKvH02].

7.4 Preparing the Data

Raw data underwent several preprocessing steps:

- **Non-respondent Removal:** The initial networks had 123 nodes (all researchers). Non-respondents were removed, reducing the number of nodes to the **79 respondents**. This also removed edges connected to non-respondents, even if mentioned by a respondent.
- **Directionality Removal:** Directionality was removed from both survey networks and the web-based network. This was done because the directionality measured by web extraction methods might not directly correspond to surveyed dimensions, and many network measures generally do not account for directionality.
- **Basic Statistics:** Table 7.1 provides statistics on the collected data before and after preprocessing, showing the number of nodes with edges and total edges for each relationship type.

7.5 Optimizing Goodness of Fit

To compare web-based networks with survey networks, the web-based network was filtered and directionality removed.

1. Filtering Parameters for Web-Based Networks Two parameters are critical for filtering the web-based network:

- **Pagecount:** The minimal number of web pages an individual must have to be included. This excludes individuals with too little web presence for effective mining.
- **Strength:** The minimal strength of relationships. This filters out ties with insufficient support. The distribution of `strength` values showed a smooth power law, while `pagecount` values had two scaling regimes (PhD students typically <1000 pages, senior members >10000 pages). Outliers due to famous namesakes (e.g., Michel Klein) were also noted.

2. Optimization Task Finding the appropriate filtering parameters is an **optimization task** aimed at maximizing the **similarity** between the survey networks and the extracted network. This involves choosing a similarity measure.

3. Similarity Measures for Graphs The study uses well-known measures from information retrieval to compare graphs based on their edge sets:

- **Jaccard-coefficient:** Measures the similarity between two sets by dividing the size of their intersection by the size of their union. For graphs, this is $|E1 \cap E2| / (|E1| + |E2| - |E1 \cap E2|)$ [35, BEF02].
- **Precision:** The proportion of extracted (predicted) edges that are correct (present in the survey network). $|E1 \cap E2| / |E1|$ [35, vR79].
- **Recall:** The proportion of correct (survey) edges that were successfully extracted. $|E1 \cap E2| / |E2|$ [35, vR79].
- **F-measure:** The harmonic mean of precision and recall, providing a single score that balances both [35, vR79]. $2 * (|E1 \cap E2|)^2 / (|E1| * |E1 \cap E2| + |E2| * |E1 \cap E2|)$.

4. Visualizing Parameter Effects Surface plots (Figures 7.3, 7.4, 7.5) visualize how changing `pagecount` and `strength` thresholds affects similarity measures.

- **F-measure (Symmetric):** Shows a convex surface with a single highest peak (optimum) and potentially a second peak representing a different trade-off between precision and recall.

- **Precision (Asymmetric):** Setting high thresholds can result in high precision (approaching 100%) but very low recall (approaching 0%), meaning only a few very certain ties are extracted.
- **Recall (Asymmetric):** Setting low thresholds (e.g., zero) can result in high recall (e.g., 82%) but low precision (e.g., 21%), meaning many ties are extracted, but many are incorrect.
- **General Finding:** It is generally easier to achieve high recall than high precision in web-based social network extraction.

5. Two-Stage Acquisition Process Suggestion Given the ease of achieving high recall, a two-stage process is suggested for large networks:

1. First, use **web mining** to collect a social network (which would have high recall).
2. Then, apply a **survey** to ask respondents to remove incorrect relations and add any missing ones. This approach reduces the burden on participants compared to traditional questionnaires by making it easier to correct a pre-selected list than to generate one from scratch.

7.6 Comparison Across Methods and Networks

The benchmark survey data also enables direct comparison of different social network mining methods.

1. Method Comparison The study compared two methods: **co-occurrence analysis** and a method based on **average precision**. Parameters for each method were chosen to optimize a specific similarity measure.

2. Results of Method Comparison

- **Average Precision Method:** Generally produces higher precision but lower recall, resulting in only slightly higher F-measure values.
- **Predictability:** Advice seeking, advice giving, and future work networks were found to be the easiest to predict for both methods, suggesting this is a feature of the data rather than the extraction method.
- **Future Work Relations:** The predictability of "future work" relations was surprising, but respondents might not have been instructed to exclude current work relations.

3. Multiplexity and p^* Analysis The relationships extracted from the Web likely reflect a combination of underlying relationships, including those surveyed and others not explicitly asked about. To measure the extent to which each surveyed relationship is present on the Web, a **p^* analysis** could be performed.

- **p* Analysis:** Assumes the web-based network is a linear combination of survey networks (Equation 7.1: $\lambda_1 S_1 + \lambda_2 S_2 + \dots = W$, where S are survey networks and W is the Web-based network). This analysis would return coefficients (λ weights) showing each survey network's contribution to the Web-based network. This could be an optimization task to maximize fit by adjusting survey network weights, rather than Web network parameters. This analysis is proposed as future work.

7.7 Predicting the Goodness of Fit

This section investigates whether certain personal attributes or Web-based indicators can predict the success of extracting an individual's social network from the Web.

1. Practical Significance Being able to predict the "goodness of fit" (similarity between an individual's real network and its Web-based image) a priori is important. If extraction is likely to fail for certain individuals, alternative methods or data sources can be used for their relationships.

2. Methodology Similarity measures are computed on **ego-networks** (the network centered around a single individual), and these are correlated with subject attributes. The mining network parameters are fixed to maximize the overall fit between survey and Web networks.

3. Attributes Considered

- **Survey-based attributes:**
 - `surveydegree`: Number of relations mentioned by the respondent.
 - `age`: Age of the individual.
 - `entry`: Years spent at the Vrije Universiteit (VU).
- **Web-based indicators:**
 - `miningdegree`: Number of relations extracted from the Web for someone.
 - `pagecount (recode)`: Logarithmically recoded number of web pages for someone's name.
- **Name ambiguity measures (from existing literature):** These measures estimate the web presence of a particular individual by applying clustering methods to search engine results [47, BM05, BMI06].
 - **NC1 (Jaccard-coefficient between first and last name):** Predicts how much a first and last name "belong together." Higher values suggest a less common name.
 - **NC2 (Ratio of pages for full name + "Vrije Universiteit" to pages for full name only):** Measures the proportion of pages

certainly about the correct individual versus total pages. Higher values indicate easier retrieval of the correct individual.

4. Findings on Attribute Correlations

- **surveydegree and miningdegree:**
 - The more respondents are mentioned by someone (`surveydegree`), the higher the **precision** of extraction. This could be because more cooperative respondents provide more relations, leading to higher precision, or because individuals with larger real networks also have a larger web presence, making extraction easier.
 - Trivially, the more relations extracted from the Web for an individual (`miningdegree`), the lower the precision and the higher the recall.
- **Survey Attributes (age, entry):** None of these had a direct influence on the result, although `age` and `entry` were correlated with `pagecount` as expected.
- **Name Ambiguity Measures:**
 - **NC1:** Had no significant effect. A problem is that it assigns high scores to common first and last names even if their combination is rare, and it cannot distinguish between name commonality (e.g., Li Ding) and famous namesakes (e.g., George Bush).
 - **NC2:** Surprisingly, had a negative effect on the F-measure. This might be because respondents with a high percentage of mentions linked to the Vrije Universiteit produce higher recall, which, at an already high recall level, can lead to lower precision.
- **Correlation Table (Figure 7.7):** Provides detailed Pearson correlation coefficients and significance levels between all attributes and similarity measures.

7.8 Evaluation Through Analysis

The ultimate goal is to assess to what extent a web-based network can replace a survey network when used in network analysis.

1. Robustness of Network Measures

- While a 100% match between survey and web-based networks is unlikely, it's not required for obtaining relevant analysis results.
- Most network measures are statistical aggregates and are relatively **robust to missing or incorrect information**.
- **Group-level analysis** (e.g., identifying relationships between sections) is typically insensitive to errors in extracting specific individual cases.

- Visual comparison (Figures 7.8 and 7.9) of the advice seeking network from the survey and the optimized mining network showed they reveal the same underlying organizational structure: the AI, BI, and Computer Systems sections formed close, densely linked relationships, while other groups were smaller or loosely connected. This was attributed to shared research interests (like Semantic Web).

2. Robustness of Centrality Measures The study tested the robustness of **centrality measures**, which identify important nodes in a network:

- **Degree Centrality:** Number of direct connections a node has.
- **Closeness Centrality:** How close a node is to all other nodes.
- **Betweenness Centrality:** How often a node lies on the shortest path between other nodes.
- **Findings:**
 - Overlap in the top 20 nodes ranked by these measures between survey and web networks: 55% for degree, 65% for closeness, and 50% for betweenness.
 - Correlation between the values of these measures: 0.67 for degree, 0.49 for closeness, and 0.22 for betweenness.
 - The higher correlation for degree centrality is explained by the optimization process calibrating networks on precision/recall, which relates to the number of connections.

3. General Argument for Web Data Viability The study argues that the level of similarity between survey and web-based networks is high enough that it is unlikely a network effect found with Web data would not hold if survey data were used. This is analogous to the concept of sampling in social sciences: if sampling is uniform, it's unlikely to introduce a significant spurious effect.

7.9 Discussion

The Web is rapidly becoming a crucial data source for network analysis due to its widespread adoption and transformation into a platform for socialization, collaboration, and dynamic data generation.

1. Importance of Evaluation Before Web data can be routinely analyzed, it's crucial to investigate its ability to replace traditional survey and observation methods. This chapter provides methodological guidance for such evaluation.

2. Conclusion on Web Data for Scientific Communities In the specific domain of **scientific communities**, electronic data collected from the Web is considered a "**close enough reflection**" of **real-world networks** to be used as input for network analysis, potentially replacing survey data.

3. Causes of Disagreement and Caution Disagreements between methods can stem from differences in what is measured or imperfections in either the survey or web extraction method. It is important to be cautious when considering survey data as an infallible "golden standard," as it might lead to reproducing the imperfections of the survey method itself.

4. Future Work Similar evaluations should be conducted across various network data domains to prove the robustness of these methods. For example, network mining could be applied to corporate networks and tested against existing databases of joint ventures or collaborations [62, Lem03].